AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

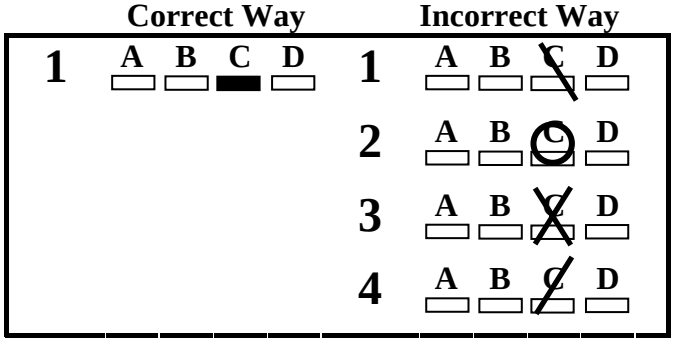
CLASS X EXAMINATION 2010

Mathematics Paper I

Time allowed: 40 minutes Marks 30

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.
6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.
7. You may use a simple calculator if you wish.

1. The square of a natural number n is equal to 4 times the previous number. Which one of the following equations correctly describes the situation?

   A. $n^2 - 4n - 4 = 0$
   B. $n^2 + 4n + 4 = 0$
   C. $n^2 + 4n - 4 = 0$
   D. $n^2 - 4n + 4 = 0$

2. The solution set of $2x - 3y + 5 = 0$ and $2x - 3y = 0$ is

   A. { }
   B. {(0, 5)}
   C. {(0, –5)}
   D. {(0, 0)}

3. Akber bought 1 litre milk and 1 dozen eggs. The price of l litre milk is Rs. 50. If he paid Rs.122, then the price of one egg is

   A. Rs. 6.0
   B. Rs. 6.8
   C. Rs. 7.0
   D. Rs. 8.5

4. The solution set of $1 + \sqrt{1 - x} = 0$ is

   A. {0}
   B. { }
   C. {1}
   D. {0, 1}

5. If $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$, then A^2

   A. is [5]
   B. is $\begin{bmatrix} 1 & 4 \end{bmatrix}$
   C. is $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$
   D. does not exist

6. If $\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \times R = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, then the order of matrix R is

A. 2×1
B. 2×2
C. 1×2
D. 1×1

7. The multiplicative inverse of $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ is

A. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
B. $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$
C. $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$
D. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

8. The multiplicative inverse of the matrix $\begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix}$ does NOT exist if

A. $x < 1$
B. $x < 0$
C. $x > 1$
D. $x = 0$

9. If $y = \sqrt{2}a$ and $z - 2\sqrt{2}a = 0$, then the relation independent of a is

A. $z - 2y = 0$
B. $z + 2y = 0$
C. $2z - y = 0$
D. $2z + y = 0$

10. If $a + x = bx$, then x is equal to

A. $\dfrac{b}{a-1}$
B. $\dfrac{b}{a+1}$
C. $\dfrac{a}{b+1}$
D. $\dfrac{a}{b-1}$

11. If a varies directly as $(b+2)$, then the constant of proportionality in terms of a and b is

A. $(b+2)$
B. $a(b+2)$
C. $\dfrac{a}{b+2}$
D. $\dfrac{b+2}{a}$

12. If $\dfrac{1}{3}a = \dfrac{10}{3}b$, then $a : b$ is

A. 1 : 3
B. 3 : 1
C. 1 : 10
D. 10 : 1

13. If $s = \dfrac{k}{\sqrt{t}}$, then the value of k according to the following table is

t	1	16
s	8	2

A. 2
B. 4
C. 6
D. 8

14. If $\dfrac{p}{q} = \dfrac{1}{2}$, then $\dfrac{p+q}{p-q}$ is equal to

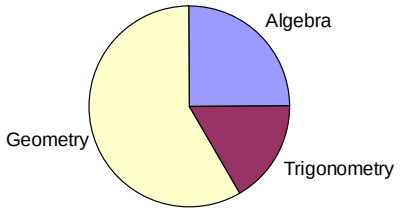
A. –3
B. 1
C. 2
D. 3

15. Which of the following is an example of discrete data?

A. Height of a student
B. Favourite colour of a student
C. Passing percentage of students
D. Number of students appeared in a test

16. For an assignment, Farzana has to solve 3 questions of algebra, 7 questions of geometry and 2 questions of trigonometry. The central angle to represent her algebra work on a pie graph is

A. 60°
B. 90°
C. 120°
D. 210°



17. The arithmetic mean of two numbers is a and the arithmetic mean of three other numbers is b. The arithmetic mean of these five numbers is

A. $\frac{2a+3b}{5}$

B. $\frac{a+b}{2}$

C. $\frac{a+b}{5}$

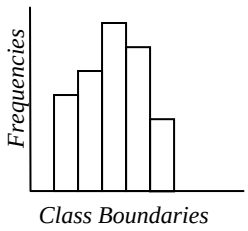
D. $\frac{2a+3b}{2}$

18. The following table gives information about number of passengers and fare paid by them on a certain day. Based on this information the median fare is

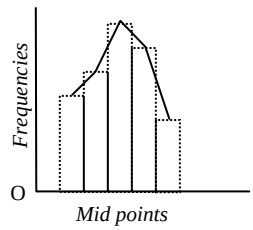
Fare in Rs.	10	26	39	48
No. of Passengers	47	165	72	34

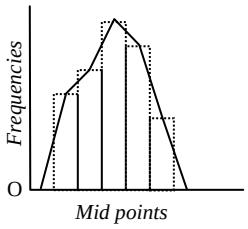
A. Rs.10
B. Rs.26
C. Rs.39
D. Rs.165

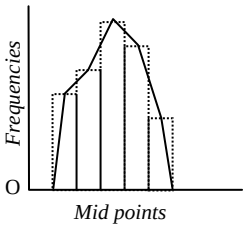
19. The diagram shows a histogram representing a frequency distribution:

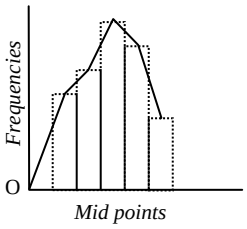


The frequency polygon associated to the above histogram is

A. 

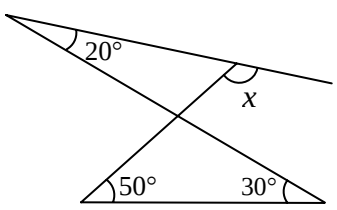
B. 

C. 

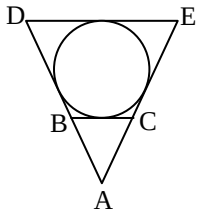
D. 

20. In the figure, the value of x is

A. 100°
B. 110°
C. 120°
D. 130°



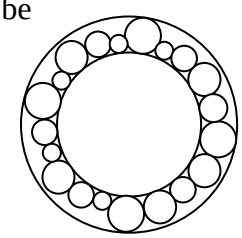
21. In the figure below $\overline{AE}, \overline{AD}, \overline{BC}$ and $\overline{DE}$ are tangents to the circle. The given circle is



I circumcircle of ΔADE
II one of the escribed circles of ΔABC
III inscribed circle of ΔADE

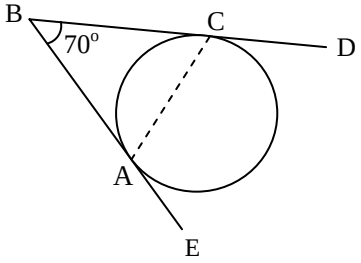
A. II and III only
B. I and II only
C. I and III only
D. I, II and III

22. Some small spheres of different diameter are fitted between two rings of diameters 20cm and 14cm. The maximum diameter of any of these spheres could be



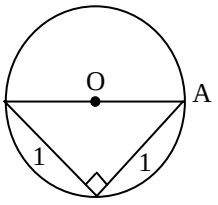
A. 3cm
B. 6cm
C. 14cm
D. 20cm

23. In the following figure BE and BD are tangents to the circle. The $m\angle BAC$ is equal to



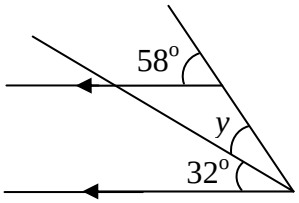
A. 20°
B. 35°
C. 55°
D. 70°

24. In the figure, radius OA of the circle is



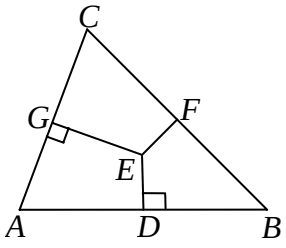
A. 1
B. $\frac{1}{\sqrt{2}}$
C. $\sqrt{2}$
D. 2

25. In the figure, the value of y is



A. 13°
B. 26°
C. 29°
D. 32°

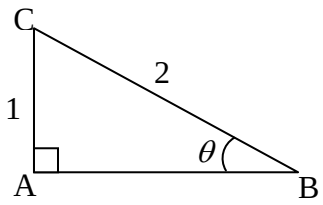
26. In the following triangle, if D, G and F are the mid points of AB, AC and BC respectively, then EF is the



A. median on BC
B. altitude on BC
C. angle bisector of angle A
D. right bisector of BC

27. In the following right angled triangle, the length of AB is

A. 1
B. $\sqrt{2}$
C. $\sqrt{3}$
D. 2

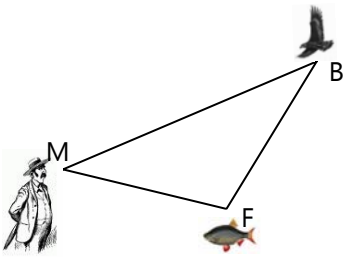


28. If in a right angled triangle, sinθ is equal to $\frac{1}{\sqrt{2}}$, then sin(90°– θ) is equal to

A. $\frac{1}{2}$
B. $\frac{1}{\sqrt{2}}$
C. $\frac{\sqrt{3}}{2}$
D. 1

29. In the following figure the angle of elevation of the man to the bird is $30°$ and the angle of depression of the bird to the fish is $65°$. The measure of $\angle MBF$ is

A. 30°
B. 35°
C. 60°
D. 65°



30. If $\sin\theta = \cos\theta$ then $\tan\theta$ is equal to

A. 0
B. $\frac{1}{\sqrt{3}}$
C. 1
D. $\sqrt{3}$

END OF PAPER

Please use this page for rough work